

2011 年考研数学三真题

一、选择题

1. 已知当 $x \rightarrow 0$ 时, 函数 $f(x) = 3\sin x - \sin 3x$ 与 cx^k 是等价无穷小, 则

A $k=1, c=4$ B $k=2, c=-4$ C $k=3, c=4$ D $k=3, c=-4$

2. 已知 $f(x)$ 在 $x=0$ 处可导, 且 $f(0)=0$, 则 $\lim_{x \rightarrow 0} \frac{x^2 f(x) - 2f(x^3)}{x^3} =$

A $-2f'(0)$ B $-f'(0)$ C $f'(0)$ D 0

3. 设 $\{U_n\}$ 是数列, 则下列命题正确的是

A 若 $\sum_{n=1}^{\infty} U_n$ 收敛, 则 $\sum_{n=1}^{\infty} (U_{2n-1} + U_{2n})$ 收敛

B 若 $\sum_{n=1}^{\infty} (U_{2n-1} + U_{2n})$ 收敛, 则 $\sum_{n=1}^{\infty} U_n$ 收敛

C 若 $\sum_{n=1}^{\infty} U_n$ 收敛, 则 $\sum_{n=1}^{\infty} (U_{2n-1} - U_{2n})$ 收敛

D 若 $\sum_{n=1}^{\infty} (U_{2n-1} - U_{2n})$ 收敛, 则 $\sum_{n=1}^{\infty} U_n$ 收敛

4. 设 $I = \int_0^{\frac{\pi}{2}} \ln \sin x dx$, $J = \int_0^{\frac{\pi}{2}} \ln \cot x dx$, $K = \int_0^{\frac{\pi}{2}} \ln \cos x dx$ 则 I, J, K 的大小关系是

A $I < J < K$ B $I < K < J$ C $J < I < K$ D $K < J < I$

5. 设 A 为 3 阶矩阵, 将 A 的第二列加到第一列得矩阵 B , 再交换 B

的第二行与第三行得单位矩阵. 记 $P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, 则

$A =$

A $P_1 P_2$ B $P_1^{-1} P_2$ C $P_2 P_1$ D $P_2 P_1^{-1}$

6. 设 A 为 4×3 矩阵, η_1, η_2, η_3 是非齐次线性方程组 $Ax = \beta$ 的 3 个线性无关的解, k_1, k_2 为任意常数, 则 $Ax = \beta$ 的通解为

- A $\frac{\eta_2 + \eta_3}{2} + k_1(\eta_2 - \eta_1)$ B $\frac{\eta_2 - \eta_3}{2} + k_2(\eta_2 - \eta_1)$
C $\frac{\eta_2 + \eta_3}{2} + k_1(\eta_3 - \eta_1) + k_2(\eta_2 - \eta_1)$ D $\frac{\eta_2 - \eta_3}{2} + k_3(\eta_3 - \eta_1) + k_3(\eta_2 - \eta_1)$

7. 设 $F_1(x), F_2(x)$ 为两个分布函数, 其相应的概率密度 $f_1(x), f_2(x)$ 是连续函数, 则必为概率密度的是

- A $f_1(x)f_2(x)$ B $2f_2(x)F_2(x)$ C $f_1(x)F_2(x)$ D $f_1(x)F_2(x) + f_2(x)F_1(x)$

8. 设总体 X 服从参数 $\lambda (\lambda > 0)$ 泊松分布 $X_1, X_2, \dots, X_n (n \geq 2)$ 为来自总体的

简单随机样本, 则对应的统计量 $T_1 = \frac{1}{n} \sum_{i=1}^n X_i, T_2 = \frac{1}{n-1} \sum_{i=1}^{n-1} X_i + \frac{1}{n} X_n$

- A $ET_1 > ET_2, DT_1 > DT_2$ B $ET_1 > ET_2, DT_1 < DT_2$
C $ET_1 < ET_2, DT_1 > DT_2$ D $ET_1 < ET_2, DT_1 < DT_2$

二、填空题

9. 设 $f(x) = \lim_{t \rightarrow 0} x(1+3t)^{\frac{x}{t}}$, 则 $f'(x) =$ _____

10. 设函数 $z = (1 + \frac{x}{y})^y$, $dz|_{(1,0)} =$ _____

11. 曲线 $\tan(x + y + \frac{\pi}{4}) = e^y$ 在点 $(0, 0)$ 处的切线方程为 _____

12. 曲线 $y = \sqrt{x^2 - 1}$, 直线 $x=2$ 及 x 轴所围成的平面图形绕 x 轴旋转所成的旋转体的体积为 _____

13. 设二次型 $f(x_1, x_2, x_3) = x^T A x$ 的秩为 1, A 中行元素之和为 3, 则 f 在正交变换下 $x = Qy$ 的标准为 _____

14. 设二维随机变量 (X, Y) 服从 $N(\mu, \mu; \sigma^2, \sigma^2; 0)$, 则 $E(XY^2) =$ _____

三、解答题

15. 求极限 $\lim_{x \rightarrow 0} \frac{\sqrt{1+2\sin x} - x - 1}{x \ln(1+x)}$

16. 已知函数 $f(u, v)$ 具有连续的二阶偏导数, $f(1, 1) = 2$ 是 $f(u, v)$ 的极值,

$z = f[(x+y), f(x, y)]$. 求 $\frac{\partial^2 z}{\partial x \partial y} \Big|_{(1,1)}$

17. 求 $\int \frac{\arcsin \sqrt{x} + \ln x}{\sqrt{x}} dx$

18. 证明 $4 \arctan x - x + \frac{4\pi}{3} - \sqrt{3} = 0$ 恰有 2 实根。

19. $f(x)$ 在 $[0, 1]$ 有连续的导数, $f(0) = 1$, 且 $\iint_{D_t} f'(x+y) dx dy = \iint_{D_t} f'(x+y) dx dy$,

$D_t = \{(x, y) | 0 \leq y \leq t, 0 \leq x \leq t\} (0 < t \leq 1)$, 求 $f(x)$ 的表达式。

20. $\alpha_1 = (1, 0, 1)^T$, $\alpha_2 = (0, 1, 1)^T$, $\alpha_3 = (1, 3, 5)^T$ 不能由 $\beta_1 = (1, a, 1)^T$, $\beta_2 = (1, 2, 3)^T$, $\beta_3 = (1, 3, 5)^T$ 线性表出, ①求 a ; ②将 $\beta_1, \beta_2, \beta_3$ 由 $\alpha_1, \alpha_2, \alpha_3$ 线性表出。

21. A 为三阶实矩阵, $R(A) = 2$, 且 $A \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ -11 \end{pmatrix} = \begin{pmatrix} -11 \\ 0 & 0 \\ 1 & 1 \end{pmatrix}$

(1) 求 A 的特征值与特征向量; (2) 求 A 。

22.

X	0	1
P	1/3	2/3

Y	-1	0	1
P	1/3	1/3	1/3

$P(X^2 = Y^2) = 1$

求: (1) (X, Y) 的分布; (2) $Z = XY$ 的分布; (3) ρ_{XY}

23. (X, Y) 在 G 上服从均匀分布, G 由 $x-y=0$, $x+y=2$ 与 $y=0$ 围成。

(1) 求边缘密度 $f_X(x)$; (2) 求 $f_{X|Y}(x|y)$ 。

答案:

一选择题

CBAEDCDD

二填空题

$$9. e^{3x}(1+3x) \quad 10. (\ln 2 + \frac{1}{2})dx - (\ln 2 + \frac{1}{2})dy \quad 11. 2x+y=0$$

$$12. \frac{4}{3}\pi \quad 13. 3y_1^2 \quad 14. \mu(\sigma^2 + \mu^2)$$

三解答题

$$15. \text{解: 原式} = \lim_{x \rightarrow 0} \frac{\frac{\cos x}{\sqrt{1+2\sin x}} - 1}{2x} = \frac{\cos x - \sqrt{1+2\sin x}}{2x} = -\frac{1}{2}$$

$$16. \text{解: } \frac{\partial z}{\partial x} = f_1'[(x+y), f(x, y)] + f_2'[(x+y), f(x, y)]f_1'(x, y)$$

$$\begin{aligned} \frac{\partial^2 x}{\partial x \partial y} &= f_{1,1}''[(x+y), f(x, y)] + f_{1,2}''[(x+y), f(x, y)]f_2'(x, y) + f_{2,1}''[(x+y), f(x, y)]f_1'(x, y) \\ &+ f_{2,2}''[(x+y), f(x, y)]f_2'(x, y)f_1'(x, y) + f_2'[(x+y), f(x, y)]f_{1,2}''(x, y) \end{aligned}$$

由于 $f(1,1)=2$ 是 $f(u,v)$ 的极值, 可知 $f_1'(1,1)=f_2'(1,1)=0$

$$\text{故 } \frac{\partial^2 z}{\partial x \partial y} \Big|_{(1,1)} = f_{1,1}''[2, f(1,1)] + f_2'[2, f(1,1)]f_{1,2}'(1,1) = f_{1,1}''(2,2) + f_2'(2,2)f_{1,2}'(1,1)$$

20.解:

$$1) \because |\alpha_1, \alpha_2, \alpha_3| = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 1 & 1 & 5 \end{vmatrix} = 1 \neq 0 \quad \therefore r(\alpha_1, \alpha_2, \alpha_3) = 3$$

又 $\because \alpha_1, \alpha_2, \alpha_3$ 不能由 $\beta_1, \beta_2, \beta_3$ 线性表示, $\therefore r(\beta_1, \beta_2, \beta_3) < 3$, 于是 $|\beta_1, \beta_2, \beta_3| = 0$, 解得 $a = 1$

$$2) (\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3) = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 3 & 2 & 3 \\ 1 & 1 & 5 & 3 & 5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 3 & 2 & 3 \\ 0 & 1 & 4 & 2 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 3 & 2 & 3 \\ 6 & 0 & 1 & -1 & 0 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 1 & -1 & 0 \end{pmatrix} \text{ 于是 } \begin{cases} \beta_1 = 2\alpha_1 + 4\alpha_2 - \alpha_3 \\ \beta_2 = \alpha_1 + 2\alpha_2 + 0\alpha_3 \\ \beta_3 = 0\alpha_1 + 0\alpha_2 + \alpha_3 \end{cases}$$

21.解:

1)

$$\text{令 } \alpha_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \text{ 则 } A\alpha_1 = -\alpha_1, A\alpha_2 = \alpha_2,$$

根据特征值向量的定义, A 的特征值为 $\lambda_1 = -1, \lambda_2 = 1$, 对应的线性无关的特征向量为

$$\alpha_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \because r(A) = 2 < 3, \therefore |A| = 0 \text{ 故 } \lambda_3 = 0$$

$$\text{令 } \alpha_3 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \text{ 为矩阵 } A \text{ 的相应于 } \lambda_3 = 0 \text{ 的特征向量} \because A \text{ 为实矩阵, 所以有 } \begin{cases} \alpha_1^T \alpha_3 = 0 \\ \alpha_2^T \alpha_3 = 0 \end{cases}$$

$$\text{即 } \begin{cases} x_1 - x_3 = 0 \\ x_1 + x_3 = 0 \end{cases} \text{ 解得 } \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$2) \alpha_1, \alpha_2, \alpha_3 \text{ 单位化得: } r_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, r_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, r_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \text{ 令 } Q = (r_1, r_2, r_3) = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix},$$

$$\text{则 } Q^T A Q = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \text{ 于是 } A = Q \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} Q^T = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

22.解:

$$1) P(X^2 = Y^2) = 1 \Rightarrow P(X^2 \neq Y^2) = 0, \text{即 } P(X=0, Y=1) = P(X=0, Y=-1) = P(X=1, Y=0) = 0$$

$$P(Y=1) = P(X=0, Y=1) + P(X=1, Y=1) = \frac{1}{3}$$

$$\therefore P(X=1, Y=1) = \frac{1}{3}, \text{同理如图:}$$

Y	-1	0	1	
X	0	1/3	0	1/3
0	1/3	0	1/3	1/3
1	1/3	1/3	1/3	

2) Z取值为-1, 0, 1

$$P(XY = -1) = P(X=1, Y=-1) = \frac{1}{3}, P(XY = 0) = P(X=0, Y=0) + P(X=0, Y=1)$$

$$+ P(X=0, Y=-1) + P(X=0, Y=1) = \frac{1}{3}$$

$$P(XY = 1) = P(X=1, Y=1) = \frac{1}{3}$$

Z	-1	0	1
P	1/3	1/3	1/3

$$3) EX = \frac{2}{3}, EY = 0, EXY = 0, DX = \frac{2}{9}, DY = \frac{2}{3}, \rho_{XY} = 0$$

23.解: 如图

$$S_G = 1, f_{XY}(x, y) = \begin{cases} 1 & (x, y) \in G \\ 0 & \text{其他} \end{cases}, f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy$$

$$1) x < 0 \text{ 或 } x > 2, f_X(x) = 0$$

$$0 < x \leq 1, f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \int_0^x 1 dy = x$$

$$1 < x < 2, f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \int_0^x 1 dy = 2 - x$$

$$\text{综上所述, } f_X(x) = \begin{cases} x & 0 < x \leq 1 \\ 2 - x & 1 < x < 2 \\ 0 & \text{其他} \end{cases}$$

$$2) f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx$$

$$\text{当 } y > 1 \text{ 或 } y < 0, f_Y(y) = 0; \text{ 当 } 0 \leq y \leq 1, f_Y(y) = \int_y^{2-y} 1 dx = 2 - 2y$$

$$\text{综上, } f_{X|Y}(X|Y) = \frac{f(x, y)}{f_Y(y)} = \begin{cases} \frac{1}{2-2y} & (x, y) \in G \\ 0 & (x, y) \notin G \end{cases}$$